Course Project

BUAD 345: Decision Analytics and Visualization

Summer 2026

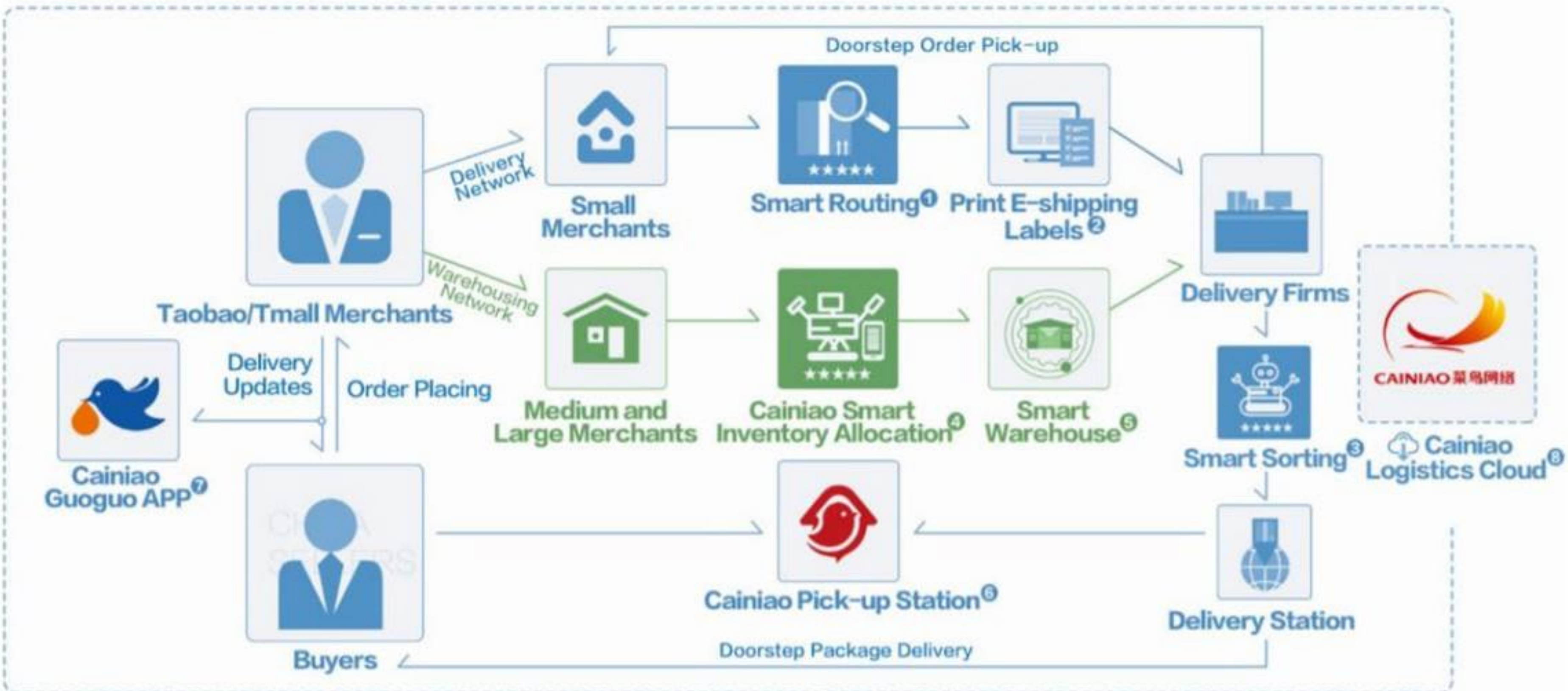
UNIVERSITY OF DELAWARE

Course Project

Part I

Background

Logistic Information Platform (LIP)



① Real-time recommendation to merchants on the best delivery option;
② E-shipping labels for easy package identification and tracking;
③ Smart sorting provides delivery firms with the best route option;
④ Based on past sales data, smart inventory allocation helps merchants stock up products in the nearest warehouse;
⑤ Cainiao partners' smart warehouses enhance efficiency in packaging and despatching;
⑥ Cainiao pick-up stations provide the last-mile delivery solution;
⑦ Cainiao Guoguo App provides one-stop order placing, package tracking and other logistics services;
⑧ Cainiao collects data throughout the logistics chain and share it with partners on its cloud platform;

Overview of Cainiao Network.
Source: Alizila

Order Fulfillment Process

1. Customer purchase an item from the Tmall marketplace
2. Seller (merchant) put delivery information into the Logistics Information Platform (LIP)
3. LIP (Cainiao) determines the optimal delivery solution
   - ❑ Assign a logistic company
   - ❑ Compute the optimal delivery route
   - ❑ Label is created in PDF format for merchant
4. Logistic company picks up the package from the seller
5. Deliver the item to the local delivery center
6. Local delivery center performs the last-mile delivery

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Part II

Data Description

Data Description

- This is a (significantly) simplified data
  - Still, a real-world dataset
- The original dataset (110 GB) was given to the members of *Manufacturing & Service Operations Management* (M&SOM) Society.
- It is not a public data. Do **NOT** share the data.

Overall Data Structure

- *CourseProject_Order.csv*
  - Order-related information (1/1/2017 – 7/31/2017)
  - 100 mb
- *CourseProject_Merchent.csv*
  - Information about merchants (sellers) (1/1/2017 – 7/31/2017)
  - 6 mb
- *CourseProject_Logistics.csv*
  - Logistics-related data (1/1/2017 – 7/31/2017)
  - 947 mb (Microsoft Excel may not be able to handle this file)
  - e.g., tracking a package

Order Data

- **day**: date of order placement (payment)
- **buyer_id**: Unique ID assigned to each buyer
- **order_id**: Unique ID assigned to each order
- **merchant_id**: Unique ID assigned to each merchant
- **item_id**: Unique ID assigned to each item
- **item_qty**: Item quantity
- **pay_amount**: Total sales price converted to USD ($)
- **pay_timestamp**: Exact time of payment
- **if_cainiao**: 1 if the order is fulfilled by Cainiao 0 otherwise
- **review**: review score on the logistic service from the customer (buyer)
- **promise**: 0 no promise; 3 three-day delivery; 2 two-day delivery; 1 one business day delivery

Order Data

	day	buyer_id	order_id	merchant_id	item_id	item_qty	pay_amount	pay_timestamp	if_cainiao	review	promise
1	2017-01-01	24403904	178077	111	381	1	88.24	2017-01-01T11:51:00Z	1	4	0
2	2017-01-01	41957800	838160	194	109904	1	13.42	2017-01-01T19:21:00Z	1	5	0
3	2017-01-01	704189	996212	32	266948	1	99.75	2017-01-01T12:01:00Z	1	5	0
4	2017-01-01	60947187	2324729	194	109904	1	13.42	2017-01-01T15:46:00Z	1	5	0
5	2017-01-01	46046490	3065588	105	176723	1	9.14	2017-01-01T13:52:00Z	1	5	0
6	2017-01-01	6877815	3151509	407	75466	1	100.90	2017-01-01T11:26:00Z	1	5	0
7	2017-01-01	39961555	3295575	42	139245	1	305.34	2017-01-01T08:51:00Z	1	NoReview	0
8	2017-01-01	68070422	3299816	42	139245	1	305.34	2017-01-01T20:04:00Z	1	5	0
9	2017-01-01	66462289	4033477	105	109360	1	7.15	2017-01-01T15:00:00Z	1	NoReview	0
10	2017-01-01	73868069	5678468	42	245825	2	59.69	2017-01-01T19:03:00Z	1	5	0
11	2017-01-01	26183003	6504660	105	176723	1	9.14	2017-01-01T11:54:00Z	1	5	0
12	2017-01-01	38435391	6972978	42	139245	1	305.34	2017-01-01T20:26:00Z	1	5	0
13	2017-01-01	1958654	7272330	105	176723	1	9.14	2017-01-01T18:12:00Z	1	NoReview	0
14	2017-01-01	23483330	8000520	105	176723	1	7.08	2017-01-01T10:40:00Z	1	5	0

Merchant Data

- **day**: date of data entry
- **merchant_id**: Unique ID for each merchant
- **pcuv**: Number of unique visitors (PC) on that day
- **appuv**: Number of unique visitors (Smartphone App) on that day
- **avgLogisticScore**: Average logistics score computed based on all scores received up to now
- **avgServiceQualityScore**: Average service quality score computed based on all scores up to now

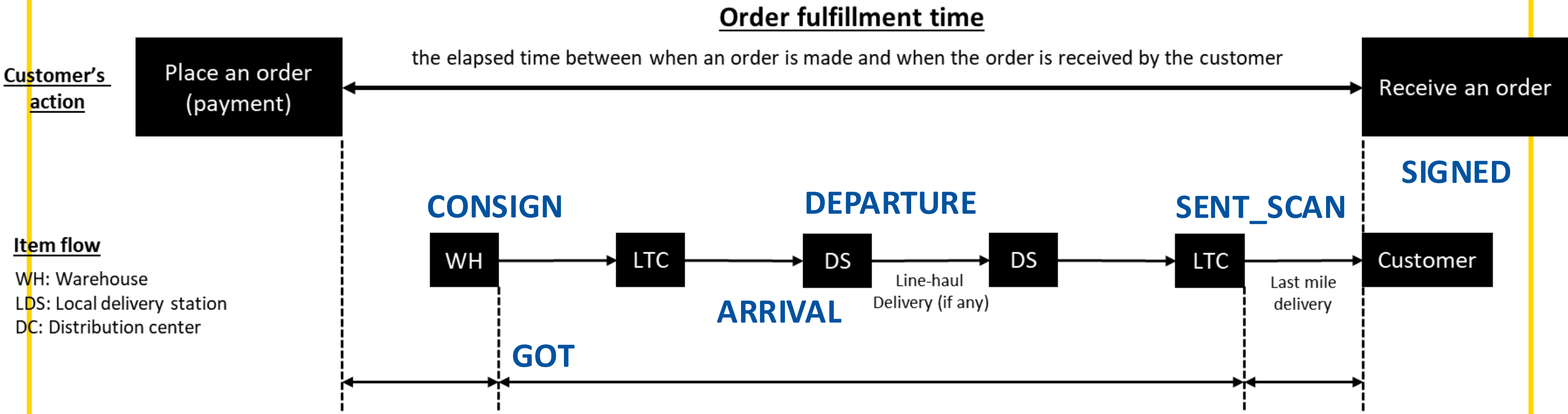
Merchant Data

	day	merchant_id	pcuv	appuv	avgLogisticScore	avgServiceQualityScore
1	20170101	1	871	10813	4.767491	4.784556
2	20170102	1	1021	12207	4.775739	4.794088
3	20170103	1	1322	12139	4.878516	4.917630
4	20170104	1	1308	13562	4.613960	4.809117
5	20170105	1	1468	19092	4.688087	4.703345
6	20170106	1	1476	19908	4.600275	4.786401
7	20170107	1	1380	14087	4.786047	4.762791
8	20170108	1	1369	16744	4.792129	4.751469
9	20170109	1	1942	17287	4.851626	4.880081
10	20170110	1	1840	14734	4.847928	4.864503
11	20170111	1	1782	13971	4.690018	4.688645
12	20170112	1	1514	12663	4.551825	4.580216
13	20170113	1	1323	12222	4.517125	4.756347
14	20170114	1	1100	11512	4.766306	4.824067
15	20170115	1	955	12246	4.614394	4.639015
16	20170116	1	1150	16419	4.443641	4.635261

Logistics Data

- **order_id**: Unique ID for each order
- **day**: date of order placement
- **logistic_order_id**: Unique ID for each order fulfillment process
- **facility_id**: Unique ID for each logistics facility
- **facility_type**: 1 small delivery station; 2 large logistic transfer center
- **city_id**: Unique ID for each city
- **logistic_company_id**: Unique ID for each logistic company (carrier)
- **timestamp**: Exact timestamp of each data entry
- **action**: Please see below
  - *CONSIGN*: dispatch of the package from warehouse/merchant
  - *GOT*: package received by the carrier
  - *ARRIVAL*: arrival at the facility
  - *DEPARTURE*: leave the facility
  - *SENT_SCAN*: package left the final delivery station for the last-mile delivery
  - *SIGNED*: package successfully received by the buyer

Logistics Data



Total order fulfillment time = **SIGNED** timestamp – **pay_timestamp**

Logistics Data

	order_id	day	logistic_order_id	action	facility_id	facility_type	city_id	logistic_company_id	timestamp
1	142	20170726	29166332	CONSIGN		NA		431	2017-07-26 13:47:00
2	142	20170726	29166332	GOT	36972	1	45	431	2017-07-26 19:46:00
3	142	20170726	29166332	ARRIVAL	21624	2	45	431	2017-07-27 03:36:00
4	142	20170726	29166332	DEPARTURE	21624	2	45	431	2017-07-27 03:37:00
5	142	20170726	29166332	ARRIVAL	122201	2	53	431	2017-07-28 14:59:00
6	142	20170726	29166332	DEPARTURE	122201	2	53	431	2017-07-28 15:00:00
7	142	20170726	29166332	ARRIVAL	75162	2	228	431	2017-07-28 20:35:00
8	142	20170726	29166332	DEPARTURE	75162	2	228	431	2017-07-29 00:09:00
9	142	20170726	29166332	ARRIVAL	182671	1	228	431	2017-07-29 07:27:00
10	142	20170726	29166332	SENT_SCAN	182671	1	228	431	2017-07-29 08:44:00
11	142	20170726	29166332	SIGNED	182671	1	228	431	2017-07-29 12:51:00
12	327	20170322	80482145	CONSIGN		NA		143	2017-03-22 13:06:00
13	327	20170322	80482145	GOT	61452	1	246	143	2017-03-22 21:03:00
14	327	20170322	80482145	ARRIVAL	170424	2	246	143	2017-03-22 22:07:00

CONSIGN -> GOT -> DEPARTURE -> ARRIVAL -> … Package is delivered to a LC's facility

CONSIGN -> GOT -> ARRIVAL -> DEPARTRUE -> … Package is picked up by the LC

Part III

Preliminary Analysis Demonstration

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Part IV

Guidelines

Workflow

- **Empirical approach (traditional approach)**
  - Literature review
  - Hypothesis development (problem identification)
  - Data collection (or finding relevant dataset)
  - Analysis and summary of results

- **Data-driven method**
  - Domain knowledge + Exploratory analysis on given data
  - Problem identification (interesting observation)
  - Further analysis (in-depth analysis)
  - summary of results & practical implication

Data-driven Method

- Who is your audience?
  - Cainiao? Merchants? Logistic companies? Customers?
- Exploratory analysis
  - Find interesting and important preliminary results?
  - Beginning of your story development (motivation)
- How to add depth to your analysis
  - Consider other variables to refine your analysis
- So what?
  - Practical implication

Who is your audience?

- Tmall.com (e-marketplace)
- Cainiao (Logistics arm)
- Merchant (sellers using Tmall.com)
- Logistic Company (carrier)
- Customer (buyers)

Exploratory Analysis

- **<u>Parallel</u>** analysis
  - You can investigate as many questions as you want
  - Domain knowledge is important!
    - Reason for your analysis
      - Does pay amount affect facility type?
    - Be aware of the *causality* issue
      - Popular products are cheap vs. Cheap products are popular ??
  - Know your data (especially the limitations)

In-depth Analysis

- You should have a big theme
  - The main objective must be something important
    - Something your audience will agree on
  - The overarching objective should reflect your interest. For example:
    - Cainiao is fast in general
    - Large logistic companies show higher customer review
    - Popular merchants tend to depend more on Cainiao
  - Now, what you should focus on is "BUT"
    - This is going to be the punchline! Example:
      - Cainiao is fast in general BUT …
      - Large LCs show higher customer review BUT …
- Further analysis should be "**hierarchical**"

Hierarchical Analysis (digging)

- *"Cainiao is fast in general"* …. But?
- Is this an apple-to-apple comparison?
  - Example: Promised speed? travel distance? Item type?
- What are the other variables that might have potential impact on delivery speed?
  - Logistic company? Merchant? Customer location? Warehouse location?
- Prioritize your secondary variables
  - Construct a hierarchy

Hierarchical Analysis (digging)

- Control (for a fair comparison)
  - Did Cainiao handle all product types? If not, should we include them in our comparison?
  - Should we exclude orders with promised speed?
- Interaction (for exciting results)
  - This analysis needs a hierarchical structure
    - Cainiao effect + LC size (large/small LC)
    - Let's say we have found that Cainiao does not perform well particularly with small LC. Why? Is there any variable in our dataset that can shed some light on this observation?
    - What about avg. # of cities passed? Maybe, small LC tends to make many stops for consolidating their packages.

Hierarchical Analysis (digging)

- Difference btw a parallel and hierarchical analysis?
  - Parallel Example
    - Delivery time comparison btw Cainiao and non-Cainiao
    - Delivery time comparison btw large LC and small LC
    - Delivery time comparison btw small endCity and large endCity
    - **Outcome**: three un-related conclusions (shallow insight)
  - Hierarchical Example
    - Delivery time comparison btw Cainiao and non-Cainiao
    - Further time comparison considering LC size
    - Based on the result above, we may add end city size
    - **Outcome**: *e.g.,* Cainiao is fast in general but not so much when the destination was a small city and the order was assigned to a small LC. (You should further discuss possible explanations for this conclusion based on domain knowledge)

So what? (Adding VALUE)

- Come back to domain knowledge
- This is very important in business analytics
  - Statistics focuses on math and inference methods
  - Applied Statistics focuses on modeling and interpretation
  - CS emphasizes algorithmic development (new/improved)
  - Business analytics is all about practical implication
- Why your results should be interesting to someone?
- I see your results. So, what’s next?

Report Structure

- <u>Motivation</u>: Delivery speed is important (needs justification from data, news article, or literature)
- <u>Main idea</u>: Let's see if Cainiao can help with making the overall delivery fast.
- <u>Your findings</u>: From our results, it seems that Cainiao is fast in general but not so much when the destination was a small city and the order was assigned to a small LC.
- <u>Potential reasons</u>: According to xxx, small cities in China have limited infrastructure compared to big cities. This increases the last-mile delivery by *x*% (based on the data). Furthermore, our data shows that small LCs take longer time (*x* hrs more on average) for order pickup (pre-delivery time).

Report Structure

– Practical implication: Based on our analysis,
  - Cainiao should focus on strengthening their business relationship with major carriers for their reliable performance. However, as discussed in recent article (*xxxx et al. 2019*), this business strategy may introduce some issues.
  - It seems that small LCs do not have enough resources to follow the optimal routing provided by the Cainiao's smart routing algorithm; hence, it might be useful to redesign the algorithm to consider the LC's capability (size, geographical coverage, experience, …). Although it was not rigorously tested in e-commerce setting, *yyyy et al. 2018* showed that such tailored algorithms can significantly improve the outcome.
  - To improve the last-mile delivery especially at small cities, Cainiao may want to consider providing drop boxes to those areas so that customers can pick up their package conveniently. According to *zzzz et al. 2000*, this potentially can reduce the last-mile delivery up to *x*%.

Remember!

- **Central Theme!!!**
  - It is all about story-telling. Focus on something specific! Do not try to overachieve.
    - We focus on customer review score, delivery speed, on-time delivery rate for promised orders, and comparison btw app and PC. …. *This is not a project with clear "focus".*
- Technical skill <<< **Good storytelling**
- Summarize your project in **two** sentences
  - Sentence 1: Overall findings (might be obvious)
  - Sentence 2: Your punchline (the uniqueness of your study)